



mkat

**PETROLEUM SAFETY TRAINING
APPLICATION**

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Introduction

This documents the proposed Petroleum Safety Training Application. This application can be used to train and update the training of people in the petroleum industry to ensure that procedures are followed and protocols are observed to ensure the safety of everyone involved. Also described is the potential integration of this application with the Job Ticketing and Site Service Management Applications for ensuring that technicians who are sent to sites for servicing possess the requisite training to fulfil the job safely.

Petroleum Safety Training Application

This application will consist of multiple courses aimed at different fields and careers within the petroleum industry. Users can be assigned quizzes when they need to complete a job or on a set schedule to ensure that procedures are reinforced. Safety courses can also be re-allocated when new content is added to keep up with industry trends and introduce new policies in reaction to any reported incidents. When training is required, users can be assigned a reference Number to enter training with (Figure 1).

A screenshot of a web application interface for 'Petrol Attendant Safety Training'. The title is centered at the top in a large, bold, black font. Below the title, the text 'Reference Number' is displayed in a smaller, bold, black font. Underneath this is a white rectangular input field with a thin grey border, containing the placeholder text 'Enter your reference number' in a light grey font. At the bottom of the form is a large, solid black rectangular button with the text 'Access Training' in a white, bold, sans-serif font. The entire form is set against a light grey background with rounded corners and a subtle drop shadow.

Figure 1 - Safety Training Login



Petrol Station Safety Training

[Take Quiz](#)

Module 1: Basic Safety Protocols

⚠ No Smoking Zone

Strictly enforce no smoking within 15 meters of fuel pumps. Display clear signage in English and local languages.

Emergency Procedures:

Fire Emergency

1. Hit emergency stop button
2. Call emergency services (10177)
3. Use fire extinguisher if safe

Fuel Spill

1. Stop fuel flow immediately
2. Apply spill kit
3. Clear area of customers

Figure 2 - Learning Content before Quiz

Users will have modules within a course which cover safety-related topics and practices. Figure 2 provides an example of an Introductory Module which reinforces the emergency contacts and practices which are applicable to all people who enter a Petroleum Industry-related site. Videos can also be made to better illustrate concepts and principles. These videos will not allow for skipping to ensure that users watch the entire video.

Once the user feels comfortable with the material, they can take a quiz to evaluate whether they have absorbed the materials presented in the module. An example of such a quiz can be observed in Figure 3. Once the user has completed the quiz, it can be submitted for assessment. This assessment can be observed in Figure 4. The Feedback will say where the user was correct and where they answered incorrectly. Any incorrect answers will have a correction to help the user learn from their mistakes. Users will only be allowed to progress to the next milestone when all questions are correctly answered.



Safety Knowledge Assessment

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What is the minimum safe distance for smoking from fuel pumps?

- ☐ 5 meters
- ☐ 10 meters
- ☐ 15 meters
- ☐ 20 meters

What is the emergency services number in South Africa?

- ☐ 10111
- ☐ 10177
- ☐ 112
- ☐ 107

What is the first step in case of a fuel spill?

- ☐ Call emergency services
- ☐ Apply spill kit
- ☐ Clear area of customers
- ☐ Stop fuel flow immediately

In case of fire, what is the correct order of actions?

- ☐ Use fire extinguisher, call emergency services, hit emergency stop
- ☐ Hit emergency stop, call emergency services, use fire extinguisher
- ☐ Call emergency services, hit emergency stop, use fire extinguisher
- ☐ Clear the area, call supervisor, use fire extinguisher

Submit Quiz

Figure 3 - Safety Quiz



Safety Knowledge Assessment

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What is the minimum safe distance for smoking from fuel pumps?

- ☐ 5 meters
- ☐ 10 meters
- ☒ 15 meters ✓
- ☐ 20 meters

What is the emergency services number in South Africa?

- ☐ 10111
- ☒ 10177 ✓
- ☐ 112
- ☐ 107

What is the first step in case of a fuel spill?

- ☐ Call emergency services ✗
- ☐ Apply spill kit
- ☐ Clear area of customers
- ☐ Stop fuel flow immediately

Correct answer: Stop fuel flow immediately

In case of fire, what is the correct order of actions?

- ☐ Use fire extinguisher, call emergency services, hit emergency stop
- ☐ Hit emergency stop, call emergency services, use fire extinguisher
- ☐ Call emergency services, hit emergency stop, use fire extinguisher ✗
- ☐ Clear the area, call supervisor, use fire extinguisher

Correct answer: Hit emergency stop, call emergency services, use fire extinguisher

Your Score: 2 / 4

Review the material and try again to improve your score.

Figure 4 - Quiz Results



Integration of this Application with the Job Ticketing and Site Service Management Applications

MKat offers API integration between the Petroleum Safety Training App and the Job Ticketing and Site Service Management Applications. What this means is that the two applications can talk to each other. Why might this be useful? Imagine a coordinator needs to send technicians to a site to react to a service request. By integrating these apps, the safety training badges of the technicians can be viewed to ensure that the technicians have the requisite training to ensure that full safety measures are observed for the required service. Likewise, it ensures that everyone involved in the daily running of the sites has undergone the appropriate training to ensure the safety of the site for all staff and clients.



MKAT

MKat specialises in data. We build the interfaces and databases that help you create, capture and store that information, not just so that's its safe, but so that you can leverage it. Data can reveal inefficiencies waiting to be optimised, trends waiting to be capitalised and insights that can be actioned.

If you are interested in any of our solutions and offerings, feel free to contact us.



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